Year 13 Maths - Hypothesis testing

Similar questions - Testing the mean of a normal distribution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A bottling machine is set to fill bottles with a mean volume of 12.4 cl, with standard deviation 0.8 cl. Following a repair, the manager believes the mean volume has decreased. A random sample of 40 bottles has mean volume 12.2 cl. Assuming the standard deviation is unchanged, test the manager's belief at the 5% significance level. **[7]**

2 The times, in seconds, taken by athletes to complete a drill are normally distributed with mean 250 and standard deviation 6. After a change of training programme, a coach takes a random sample of 25 athletes and finds a mean time of 252.9 s. The standard deviation is assumed unchanged.

(a) Write down suitable hypotheses to test whether the mean time has changed. **[2]**

(b) Find the critical region for the sample mean at the 5% significance level. **[4]**

(c) Complete the test and state your conclusion in context. **[2]**

Question 1

$H_0 : \mu = 12.4$, $H_1 : \mu < 12.4$ (one-tailed). Under H_0, $\bar{X} \sim N\left(12.4, \frac{0.8^2}{40}\right)$. $z = \frac{12.2 - 12.4}{0.8/\sqrt{40}} = -1.581$ and $P(\bar{X} \leqslant 12.2) = 0.0569 > 0.05$ (critical $z = -1.645$). Do not reject H_0: there is insufficient evidence at the 5% level that the mean volume has decreased

Question 2

(a) $H_0 : \mu = 250$, $H_1 : \mu \neq 250$

(b) Under H_0, $\bar{X} \sim N\left(250, \frac{36}{25}\right)$, standard error 1.2. Critical values $250 \pm 1.96 \times 1.2 = 247.6$ and 252.4 (to 4 s.f.). Critical region: $\bar{x} < 247.6$ or $\bar{x} > 252.4$

(c) $252.9 > 252.4$ is in the critical region (equivalently $z = 2.417 > 1.96$; p-value $0.0157 < 0.05$), so reject H_0: there is evidence at the 5% level that the mean time has changed (it has increased)